Nikhilesh Gawhale

AI Engineer | Generative AI, LLMs, RAG and Agentic Systems | Python, Cloud and Production ML
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PROFESSIONAL SUMMARY

AI Engineer with three years of professional software engineering experience, including eighteen months building and deploying production Generative AI, Large Language Model (LLM) and agentic AI systems for a regulated investment banking technology platform. Proven track record delivering end-to-end AI solutions: Retrieval-Augmented Generation (RAG) pipelines, multi-agent orchestration, prompt engineering, model fine-tuning, and the evaluation frameworks, guardrails and monitoring that make AI systems production-ready and auditable. Strong Python and SQL, with hands-on cloud experience across AWS and Microsoft Azure, containerisation, and continuous integration and continuous delivery (CI/CD). AI-native in working style, using coding agents including Claude Code and Cursor daily to build, test and ship software. MSc Artificial Intelligence with Distinction from Queen Mary University of London. Comfortable working independently with minimal supervision, translating ambiguous business problems into scoped, production-grade technical solutions, and communicating clearly with both technical and non-technical stakeholders.

TECHNICAL SKILLS

Generative AI and Large Language Models (LLMs): Anthropic Claude, OpenAI GPT, Azure OpenAI, AWS Bedrock, Google Gemini (API-level); prompt engineering, prompt optimisation, structured outputs, function calling and tool use; LoRA fine-tuning of language models on a 10,000-document enterprise corpus; context engineering and context-window management
Agentic AI and Multi-Agent Systems: Designing and building AI agents, agentic workflows and multi-agent orchestration; LangChain, LangGraph, CrewAI, AutoGen; Model Context Protocol (MCP) servers and tool integrations; autonomous task decomposition, planning, memory, retries, fallbacks and human-in-the-loop approval
Retrieval-Augmented Generation (RAG) and Vector Databases: End-to-end RAG architecture including data ingestion, chunking strategy, embeddings, hybrid dense and sparse retrieval, and re-ranking; vector databases including pgvector, Milvus, Azure AI Search (Pinecone, Weaviate, FAISS - equivalent architecture); semantic search and knowledge retrieval from structured and unstructured data
AI Evaluation, Observability and Governance: LLM evaluation frameworks with labelled gold sets, per-class precision, recall and F1, LLM-judge scoring and calibration; continuous integration (CI) regression testing; tracing, logging and monitoring across LLM calls, retrieval and tool invocations; guardrails, prompt-injection defence, schema-validated output, human-in-the-loop review, audit logging and Responsible AI practices
Programming, APIs and Data Engineering: Python (asyncio, FastAPI, Pydantic v2, pytest, pandas, NumPy) - strong; SQL across PostgreSQL, SQL Server and Azure SQL; JavaScript, TypeScript and Angular at working level; REST APIs, microservices, webhooks and event-driven architecture; ETL pipelines, data ingestion and data quality for AI-ready datasets; MongoDB, Redis
Machine Learning, Cloud and DevOps: PyTorch, scikit-learn, deep learning, transformers, Bayesian neural networks, classification, clustering and anomaly detection (MSc Deep Learning 90%, Machine Learning Algorithms 89%); AWS (Bedrock, Lambda, S3) and Microsoft Azure (Azure OpenAI, Azure AI Search, AKS); Docker, Kubernetes, CI/CD (GitHub Actions, GitLab CI/CD), Terraform (familiar); n8n workflow automation

PROFESSIONAL EXPERIENCE

AI Engineer, RP-Tech International Solutions (Investment Banking Technology), Remote | **January 2025 - August 2026**

- Designed, developed and deployed production-grade Generative AI and agentic AI applications for a regulated financial services platform, owning delivery end-to-end from ambiguous business problem through to production deployment and ongoing improvement.
- Architected and built Retrieval-Augmented Generation (RAG) pipelines over large volumes of unstructured documents: data ingestion, chunking strategy, embeddings, hybrid retrieval and re-ranking, with verification that outputs were grounded in and traceable to source text.
- Designed and built multi-step agentic workflows and multi-agent systems using LangGraph and LangChain, with tool calling into internal systems and databases, memory, retries, fallbacks and human-in-the-loop approval gates.
- Fine-tuned language models using LoRA on a corpus of over 10,000 financial documents to improve domain-specific performance, including dataset preparation, training and held-out evaluation.
- Built and maintained the AI evaluation framework governing production releases: a labelled benchmark of 500 documents with per-class precision, recall and F1, and automated regression testing wired into continuous integration, improving model accuracy from 70 percent to 80 percent and macro-F1 from 0.64 to 0.74.
- Implemented guardrails, validation frameworks and monitoring systems for AI applications, including prompt-injection defence, schema-validated output, scoped access controls, human review gates and immutable audit logging, satisfying compliance and Responsible AI requirements in a regulated environment.
- Built REST APIs and microservices in Python and FastAPI integrating AI capabilities with existing enterprise systems, deployed via Docker and Kubernetes with continuous integration and continuous delivery (CI/CD) across AWS and Azure, with full observability for latency, throughput and cost.
- Used AI coding agents, including Claude Code, extensively throughout development for implementation, test generation, debugging and documentation, applying critical review and quality control to all generated output.

- Worked directly with business stakeholders and domain experts to translate ambiguous requirements into scoped technical solutions, and documented reusable patterns, prompt templates and integration components adopted across later projects.

Generative AI Engineer, Midas Advisory, London, United Kingdom | **August 2024 - December 2024**

- Built Retrieval-Augmented Generation (RAG) pipelines and Generative AI features in Python and FastAPI for client-facing use cases, deployed through Docker and continuous integration and continuous delivery (CI/CD).
- Fine-tuned large language models using LoRA on a domain-specific corpus of over 10,000 financial documents, covering dataset preparation, training and evaluation against a held-out set.
- Built workflow automations using n8n, connecting multiple business tools and systems end to end for client operations.

Software Developer, Global Technologies, Pune, India | **August 2022 - September 2023**

- Built and deployed production Python services, REST APIs and machine learning data pipelines in an Agile team, with disciplined version control, automated testing and continuous integration and continuous delivery (CI/CD).
- Developed anomaly detection and predictive monitoring models over large operational datasets, improving system reliability by 45 percent.

PROJECTS

Underwriter Copilot - Agentic Document Analysis System, Python, Azure OpenAI, Azure AI Search, MongoDB, event-driven microservices | **github.com/Nikhileshgawhale/underwriter-copilot**

- Built solo, end to end, largely through AI-assisted development with Claude Code: an agentic system for insurance policy review, including clause comparison against standard wordings, deviation and risk flagging, and precedent retrieval, with every finding traced back to source text.
- Designed a gold-set evaluation framework measuring per-class F1 and retrieval precision rather than a single aggregate accuracy figure, because different categories of error carried materially different real-world costs.

Multi-Agent LLM Workflow Assistant, Python, FastAPI, Streamlit, Redis, Vector Database, Claude, OpenAI | **Personal project, 2025**

- Designed a multi-agent system in which Planner, Researcher, Coder and QA agents share Redis-backed state, retrieve context from a vector database, and pass work through a quality-assurance gate that scores output before release.

EDUCATION

MSc Artificial Intelligence - Distinction, Queen Mary University of London, United Kingdom | **September 2023 - September 2024**

Modules: Deep Learning (90%), Machine Learning Algorithms (89%), Bayesian Neural Networks (88%), Natural Language Processing (85%)

BCA Bachelor of Computer Applications - GPA 7.8/10, Shivaji Science College, Amravati, India | **August 2019 - August 2022**